



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous College Affiliated to University of Mumbai)

Re-Examination

July 2019

Max. Marks: 100

Class: SE

Course Code: IT44

Name of the Course: Operating Systems

Duration: 180 Minutes

Semester: IV

Branch: IT

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams.
- (3) Assume suitable data if necessary

Question No.		Max. Marks	CO															
Q 1 (a)	Explain 7 different functions of operating system. OR Differentiate between system call and function call	7	1															
Q 1 (b)	Explain in detail monolithic kernel architecture in detail with neat diagram.	5	1															
Q. 1(c)	Explain RAID 0 and RAID 1.	4	2															
Q.1 (d)	Given the following queue -- 95, 180, 34, 119, 11, 123, 62, 64 with the Read-write head initially at the track 50 and the tail track being at 199.Solve using FCFS. Calculate seek length and average seek length. Show with neat diagram track visited.	4	2															
Q2 (a)	What is PCB? Why it is needed? Explain PCB with neat diagram. OR Name different types of scheduler? Explain their role?	10	2															
Q2 (b)	Solve the following using FCFS and RR scheduling with time quantum 4 ms. Calculate turn around time, waiting time, average turn around time, average waiting time, completion time ? <table border="1"><thead><tr><th>Process</th><th>Arrival time</th><th>Burst time</th></tr></thead><tbody><tr><td>P1</td><td>0</td><td>8</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>2</td><td>9</td></tr><tr><td>P4</td><td>3</td><td>5</td></tr></tbody></table>	Process	Arrival time	Burst time	P1	0	8	P2	1	4	P3	2	9	P4	3	5	10	2
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Q3 (a)	Explain bankers algorithm with suitable example.	10	3
Q3 (b)	A shared variable x, initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to memory, and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution? Justify your answer. OR A solution to the Dining Philosophers Problem which avoids deadlock is -----. (a) ensure that all philosophers pick up the left fork before the right fork (b) ensure that all philosophers pick up the right fork before the left fork (c) ensure that one particular philosopher picks up the left fork before the right fork, and that all other philosophers pick up the right fork before the left fork (d) None of the above . Justify your answer	5	3
Q 3 (c)	What is deadlock. Explain necessary and sufficient conditions for deadlock.	5	3
Q4 (a)	Explain thrashing with neat diagram. Consider the following reference string 0,2,1,6,4,0,1,0,3,1,2,1. Apply the FIFO page replacement policy .Calculate miss and hit ratio for frame size 4.	10	4
Q4 (b)	Explain the working of TLB with neat diagram. Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is _____. OR A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is?	10	4



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Q5 (a)	Explain virtual memory with neat diagram.	5	4
Q 5 (c)	Explain file system software architecture	5	5
Q5 (b)	Explain Unix file system. OR Explain record blocking in detail.	10	5